Simulation of Si-Pixel vx-μSR using musrSim

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Abstract

This report documents a simplified `musrSim`/Geant4 study of a Si-pixel vertex-reconstructed μSR geometry inspired by Mandok et al. (2026). The purpose is not to reproduce the full mechanical detector module, but to test the essential ingredients of the reconstruction chain: transport of surface muons through thin silicon tracking layers, reconstruction of the incoming muon trajectory from upstream hits, extrapolation to the target plane, comparison with the true muon stopping position, and, in a second simulation, the reconstruction of a transverse-field vx-μSR time spectrum from decay positron tracks. Two configurations are considered: a zero-field track-reconstruction study with an aluminium target and variable inner-layer spacing, and a transverse-field vx-μSR study using a smaller aluminium target in a uniform magnetic field of 6.3 mT.

1 Simulation Geometry and Configuration

The simulated geometry is a simplified version of the Si-pixel vx-μSR spectrometer described by Mandok et al. (2026). The beam axis defines the z-direction, the vertical detector coordinate is y, and the target is centered at $z = 0$. The incoming particles are positive surface muons traveling along $+z$, with initial spin polarization along $+x$.

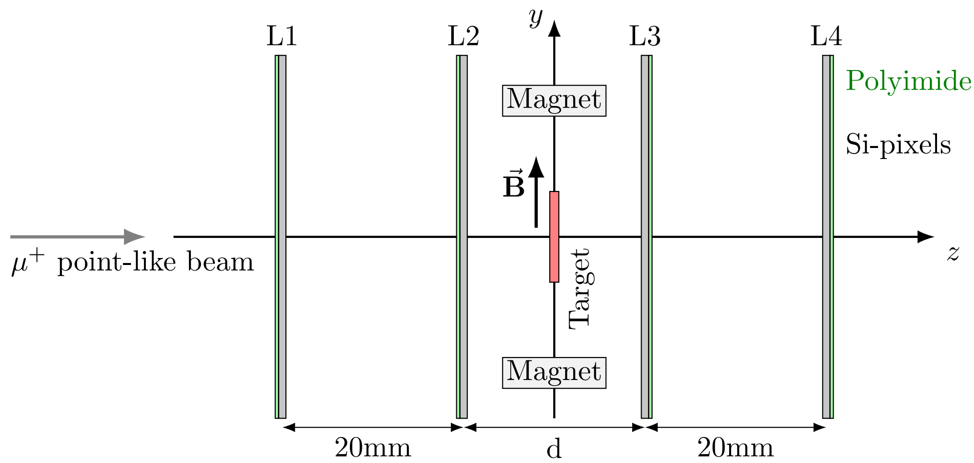


Figure 1: Schematic yz-cross section of the simplified Si-pixel vx-μSR geometry. The z-axis is the beam direction and the aluminium target is centered at $z = 0$. Layers L1–L2 reconstruct the incoming muon track, while L3–L4 are used for downstream positron tracking. In the transverse-field configuration, a uniform field is applied along $+y$. The drawing is not to scale.

The detector is approximated by four continuous silicon tracking planes rather than by the full Quad-module structure. Each silicon layer is followed by a thin polyimide/Kapton support foil. The detector-layer positions used for the reference geometry are summarized in Table 1.

The two iron magnet volumes are included only as simple geometrical placeholders. In the transverse-field simulation the relevant field is not produced by a realistic magnet model, but is

Table 1: Tracking-layer geometry used in the simplified musrSim model.

Layer	Position z [mm]	Role	Detector ID
L1	-30	upstream outer layer	101
L2	-10	upstream inner layer	102
L3	$+10$	downstream inner layer	103
L4	$+30$	downstream outer layer	104

Table 2: Main detector, target, beam, and field parameters used in the simulations.

Component	Parameter	Value / implementation
Si tracking layers	active area	40 mm $\times$ 40 mm
	thickness	100 µm
	material	`G4_Si`
Support foils	thickness	25 µm
	material	`G4_KAPTON`
Target, Simulation 1	material and size	`G4_Al`, $D = 20\,\mathrm{mm}$, $H = 1\,\mathrm{mm}$
Target, Simulation 2	material and size	`G4_Al`, $D = 6\,\mathrm{mm}$, $H = 1\,\mathrm{mm}$
Primary beam	particle	μ^+
	kinetic energy	$4.1\,\mathrm{MeV}$
	direction	$+z$
	initial polarization	$+x$, polarization fraction = 100%
Beam model	transverse profile	point-like, monoenergetic beam; no collimator
Magnetic field	Simulation 1	$B = 0$
	Simulation 2	uniform field, $\vec{B} = (0, 6.3, 0)$ mT
Magnet placeholders	material and size	`G4_Fe`, 10 mm $\times$ 2 mm $\times$ 10 mm

imposed as a uniform field in the target region. The field points along $+y$, perpendicular to both the beam direction and the initial muon spin.

The electromagnetic physics configuration includes multiple scattering, ionisation, bremsstrahlung, and pair-production processes for positive muons, the corresponding electromagnetic processes for electrons and positrons, positron annihilation, and the standard photon interactions. A user step limit of 0.01 mm is applied in the thin silicon layers and in the target in order to improve tracking through small volumes.

Software Environment

For reproducibility, all simulation macros and analysis scripts are archived in the project github repository. The simulations were performed with `musrSim` 1.0.5 using Geant4 10.3, and the ROOT output was analysed with ROOT 6.36.06, `uproot` 5.7.4, and Python 3.12.13.

2 Simulation 1: Track-Reconstruction Residual versus Inner-Layer Distance

The first simulation campaign quantifies how the track-reconstruction residual depends on the distance d between the two inner tracking layers L2 and L3. For each detector configuration, 10^5 primary μ^+ events were simulated. The magnetic field was set to zero, and the target was an aluminium disk centered at $z = 0$ with diameter $D = 20\,\mathrm{mm}$ and thickness $H = 1\,\mathrm{mm}$.

Table 3: Simulation conditions for the reconstruction-residual study.

Quantity	Value / selection
Primary particle	μ^+
Muon kinetic energy	$4.1\,\mathrm{MeV}$
Beam direction	$+z$
Initial polarization	$+x$
Magnetic field	$B = 0$
Target	`G4_Al`, $D = 20\,\mathrm{mm}$, $H = 1\,\mathrm{mm}$
Inner-layer distance	$d = 5\,\mathrm{mm}, 10\,\mathrm{mm}, \ldots, 40\,\mathrm{mm}$
Events per configuration	10^5

For each accepted event, the incoming muon trajectory is reconstructed from the hit positions in the two upstream layers L1 and L2. A straight-line tracklet is formed and extrapolated to the target plane. The extrapolated transverse position is then compared with the true simulated muon stopping position in the target. The simulation-level geometrical acceptances for muon and positron track candidates are shown as functions of the inner-layer distance d in Fig. 2.

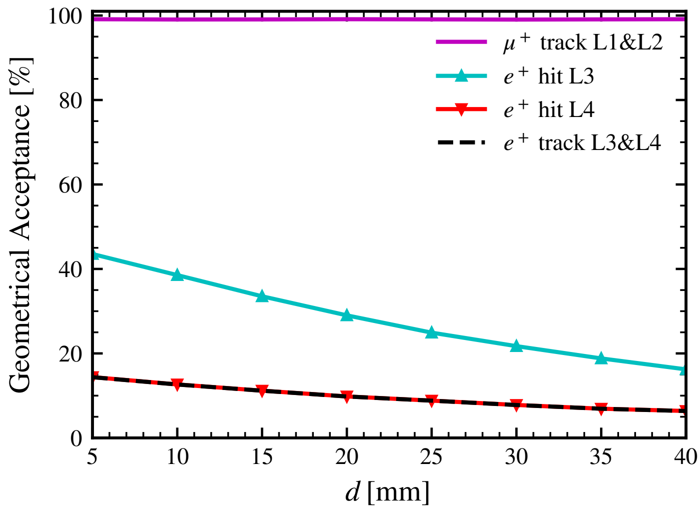


Figure 2: Simulation-level geometrical acceptance as a function of the inner-layer distance d. The μ^+ track acceptance requires valid hits in both upstream layers L1 and L2. The positron single-layer acceptances require a hit in L3 or L4 and are normalized to the number of muons decaying in the target. The e^+ track acceptance requires valid hits in both L3 and L4, with the same target-decay normalization. Since no detector-response inefficiency is included, these curves represent geometrical and analysis-selection acceptance, not experimental detector efficiency.

The upstream muon-track acceptance remains close to unity (~ 99 %) for all tested configurations, whereas the positron L3–L4 track acceptance decreases with increasing d, making downstream positron tracking the acceptance-limiting part of the reconstruction.

The muon reconstruction residual is defined as

$$\delta_\mu = \sqrt{(x_{\rm stop} - x_{\rm ext})^2 + (y_{\rm stop} - y_{\rm ext})^2},$$

where $(x_{\rm stop}, y_{\rm stop})$ is the true muon stopping position and $(x_{\rm ext}, y_{\rm ext})$ is the extrapolated position obtained from the L1–L2 tracklet. Only events with a valid target stop and a clean upstream muon track candidate are used for the δ_μ distribution.

An analogous residual is computed for decay positrons. In this case, positron tracklets are reconstructed from the downstream layers L3 and L4 and extrapolated backwards to the true muon decay position in the target. The positron residual is

$$\delta_{e^+} = \sqrt{(x_{\rm dec} - x_{\rm ext})^2 + (y_{\rm dec} - y_{\rm ext})^2}.$$

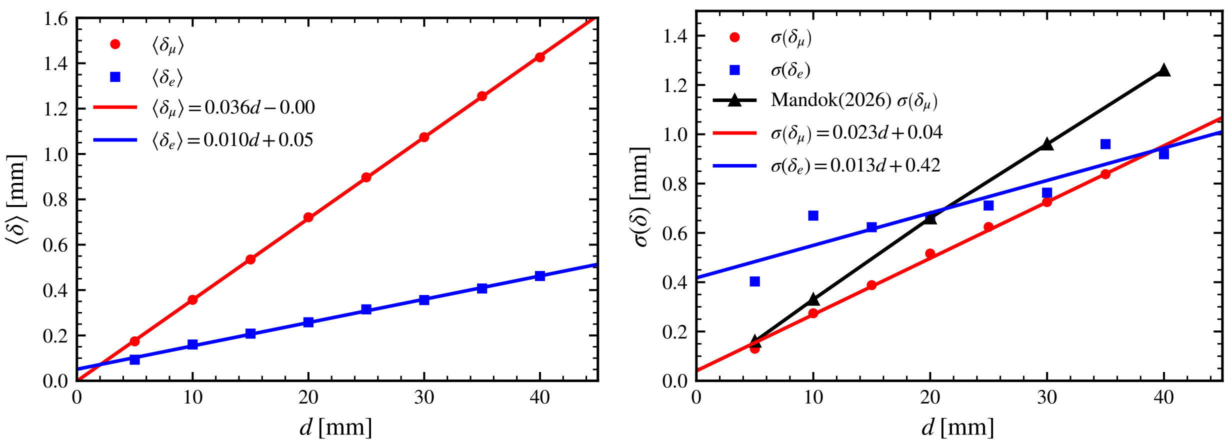
For each value of d, the distributions of δ_μ and δ_{e^+} are summarized by their mean values,

$$\langle \delta_\mu \rangle, \qquad \langle \delta_{e^+} \rangle,$$

and standard deviations,

$$\sigma(\delta_\mu), \qquad \sigma(\delta_{e^+}).$$

The standard deviation $\sigma(\delta_\mu)$ is used as the main estimate of the lateral uncertainty of the reconstructed muon stopping position. The distribution of δ_μ and δ_e for a fixed d is shown in Fig. 5 in the Appendix. The dependence of the residual mean and width on the inner-layer separation d is shown in Fig. 3.



(a) Mean residual.

(b) Residual width.

Figure 3: Mean and standard deviation of the reconstruction residuals as functions of the inner-layer distance d. The width $\sigma(\delta_\mu)$ is used as the main estimate of the muon vertex-reconstruction uncertainty.

3 Simulation 2: Transverse-Field vx-μSR Spectrum

The second simulation tests whether the simplified geometry can reproduce the basic ingredients of a transverse-field vx-μSR measurement. The detector geometry is kept fixed at the reference configuration with an inner-layer separation $d = 20\,$mm. The target is changed to a smaller aluminium disk centered at $z = 0$, with diameter $D = 6\,$mm and thickness $H = 1\,$mm. A uniform magnetic field is applied in the target region,

$$\vec{B} = \left(0,\ 6.3 \times 10^{-3}\,\mathrm{T},\ 0\right),$$

corresponding to a transverse field of $B = 6.3\,\mathrm{mT}$ along $+y$. Since the beam travels along $+z$ and the initial muon-spin polarization is along $+x$, the spin precesses around the magnetic-field direction.

Table 4: Simulation conditions for the transverse-field vx-μSR study.

Quantity	Value / implementation
Primary particle	μ^+
Muon kinetic energy	$4.1\,\mathrm{MeV}$
Target	`G4_Al`, $D = 6\,\mathrm{mm}$, $H = 1\,\mathrm{mm}$
Inner-layer distance	$d = 20\,\mathrm{mm}$
Magnetic field	$\vec{B} = (0, 6.3 \times 10^{-3}\,\mathrm{T}, 0)$
Field direction	$+y$
Initial spin direction	$+x$
Time window	$0 < \Delta t < 13\,\mu\mathrm{s}$
Events simulated	10^5

Two spectra are constructed. The first is a truth-level spectrum, where the muon decay time and positron direction are taken directly from simulated quantities. The second is a reconstructed vx-μSR spectrum. In the reconstructed case, the incoming muon track is reconstructed from L1–L2 hits, while decay-positron tracklets are reconstructed using either the upstream or downstream detector pairs. Events are accepted if the muon decays in the target, the required muon and positron tracklets are present, the reconstructed positron–muon matching distance satisfies

$$d_{\mathrm{match}} \leq 1\,\mathrm{mm},$$

and the reconstructed time difference lies inside the analysis window, begginer

$$0 < \Delta t < 13\,\mu\mathrm{s}.$$

The reconstructed time differences are defined as

$$\Delta t_{\mathrm{up}} = t^{\mathrm{rec}}_{e^+,\mathrm{up}} - t^{\mathrm{rec}}_{\mu}, \qquad \Delta t_{\mathrm{down}} = t^{\mathrm{rec}}_{e^+,\mathrm{down}} - t^{\mathrm{rec}}_{\mu}.$$

Here t^{rec}_{μ} is defined as the average hit time of the reconstructed upstream muon tracklet in L1 and L2. Similarly, $t^{\mathrm{rec}}_{e^+,\mathrm{up}}$ is the average hit time of an upstream positron tracklet reconstructed from L1 and L2, while $t^{\mathrm{rec}}_{e^+,\mathrm{down}}$ is the average hit time of a downstream positron tracklet reconstructed from L3 and L4.

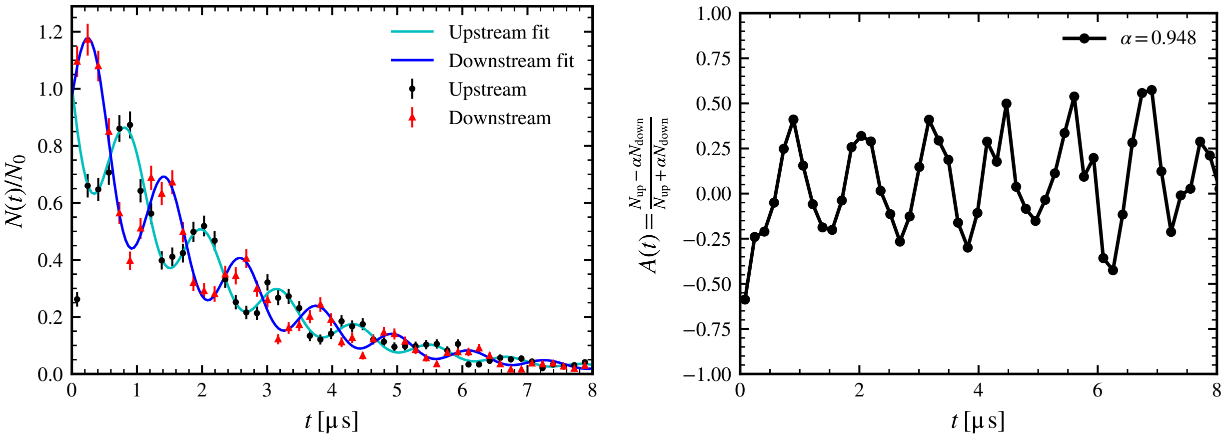
The upstream and downstream spectra are then fitted with a muon-decay envelope modulated by transverse-field spin precession,

$$N(t) = N_0 e^{-t/\tau_\mu} \left[1 + A\cos(\omega t + \phi)\right], \qquad \tau_\mu = 2.197\,\mu\mathrm{s}.$$

The asymmetry is computed from the two spectra as

$$A(t) = \frac{N_{\mathrm{up}}(t) - \alpha N_{\mathrm{down}}(t)}{N_{\mathrm{up}}(t) + \alpha N_{\mathrm{down}}(t)}, \qquad \alpha = \frac{\sum_t N_{\mathrm{up}}(t)}{\sum_t N_{\mathrm{down}}(t)}.$$

The resulting spectra and asymmetry are shown in Fig. 4. The purpose of this simulation is not yet a precision extraction of the relaxation parameters, but a consistency check that the simulated detector geometry, event matching, and time reconstruction produce the expected transverse-field oscillatory structure.



(a) Reconstructed upstream and downstream spectra.

(b) Corresponding upstream–downstream asymmetry.

Figure 4: Reconstructed transverse-field vx-μSR signal for a 6 mm-diameter aluminium target in a uniform 6.3 mT field. The spectra are built from matched muon and positron tracklets inside the time window $0 < \Delta t < 13\,\mu\text{s}$. The normalization constants N_0 are obtained from the fits and are listed in Table 5. The asymmetry shows the expected oscillatory behaviour, although the present statistics are limited.

Table 5: Reconstructed transverse-field vx-μSR spectra fit-parameters shown in Fig. 4.

Parameter	Upstream	Downstream
N_0 [counts]	389.3	377.9
A	-0.270	-0.336
ω [rad/μs]	5.373	5.381
ϕ [rad]	-1.596	1.490

Table 6: Summary of geometrical acceptance for detector configuration for transverse-field vx-μSR.

Parameter	Value	Comment
Muon L1–L2 track acceptance	99.13 %	$N_{\mu-\text{tracks}}/N_{\text{events}}$
Muon target-stop fraction	67.50 %	$N_{\mu\,\text{stop target}}/N_{\text{events}}$
Target-stop fraction after muon-track selection	68.09 %	$N_{\mu\,\text{stop target}}/N_{\mu-\text{tracks}}$
Upstream e^+ track acceptance	9.70 %	$N^{\text{up}}_{\text{e}-\text{tracks}}/N_{\mu\,\text{stop target}}$
Downstream e^+ track acceptance	10.41 %	$N^{\text{down}}_{\text{e}-\text{tracks}}/N_{\mu-\text{stop target}}$
Accepted upstream vx-μSR events	75.33 %	$N^{\text{vx}}_{\text{up}}/N^{\text{up}}_{\text{e}-\text{tracks}}$
Accepted downstream vx-μSR events	74.07 %	$N^{\text{vx}}_{\text{down}}/N^{\text{down}}_{\text{e}-\text{tracks}}$

4 Current Simplifications

The present model keeps only the ingredients needed for a first reconstruction and vx-μSR feasibility test. The real Quad-module geometry, individual chip gaps, pixelization, PCB material, detailed support structures, realistic beam spot, Pb collimation, detector timing response, time-over-threshold response, and realistic permanent-magnet field map are not included. The magnetic volumes in the macros are placeholders; the field used in the second study is a uniform field applied to the target region.

Despite these simplifications, the simulations provide a compact validation chain: first, the geometrical track-reconstruction residual is quantified as a function of detector spacing; second, the same simplified geometry is used to recover a transverse-field vx-μSR oscillation from matched muon and positron tracklets.

5 Comments on the comparison with Mandok et al.

Many results obtained in this simplified study validate those shown in the Mandok et al. For instance, the linear dependence of $\sigma(\delta_\mu)$ from d, energy deposition in the first layer L1 of μ^+ and e^+, the oscillation frequency of vx-μSR spectrum was quoted $f = 0.85$ MHz and here is $f = \frac{\omega}{2\pi} = 0.855$ MHz upstream or 0.856 MHz downstream.

There are also some disagreements, that may be worth noting. First, regarding the dependence of the $\sigma(\delta_\mu)$ as a function of d, see Fig. 3. Mandok et al. results have a steeper line in comparison to the results obtained here this is also reflected on the line gradient that for Mandok et al. is quoted ~ 0.03 while here is ~ 0.023. I wonder if the origin of this slight disagreement is anything alse then just statistics, and if it can be used to debug any potential error that might have been done here.

Second the vx-μSR spectrum near the lowest time in downstream, the $N(t)$ is smaller then it should be.

A Appendix: Sanity Checks and Supplementary Plots

A.1 Simulation 1

This appendix collects diagnostic plots used to validate the target stopping distribution (Fig. 7), energy deposition (Fig. 6), and residual distributions for selected detector spacings (Fig. 5).

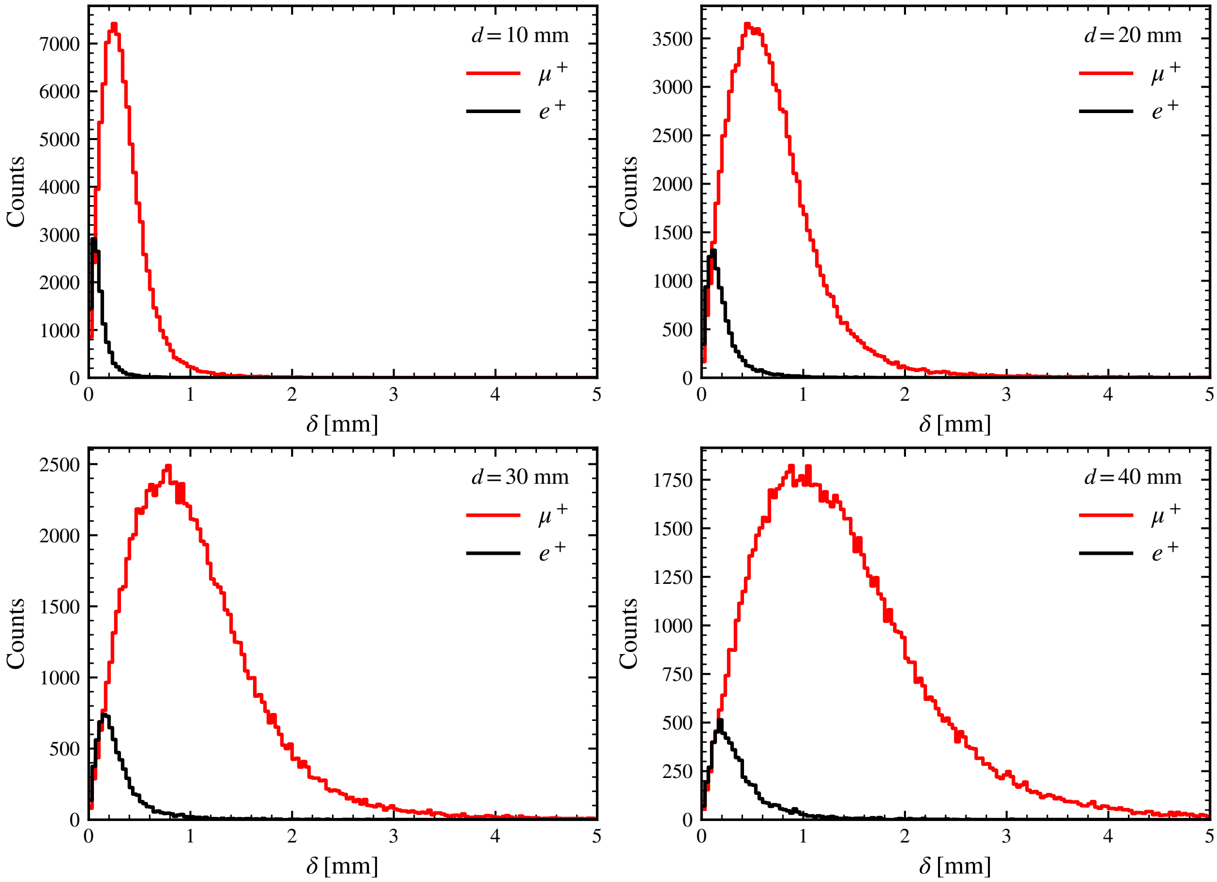


Figure 5: Distribution of δ_μ (red) and δ_e (black) for different distances d.

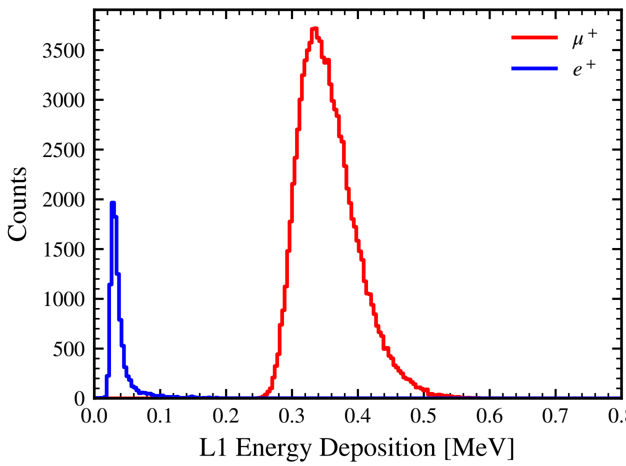


Figure 6: Energy deposit from muons and positrons in the first $L1$ upstream layer.

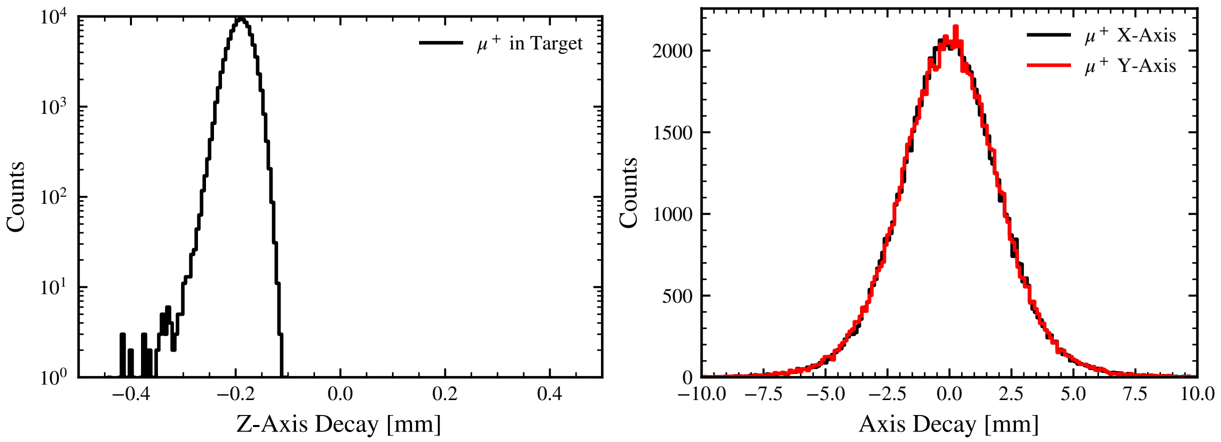


Figure 7: Muon stopping distribution in the aluminium target. Left: longitudinal distribution along z. Right: transverse distributions in x and y.

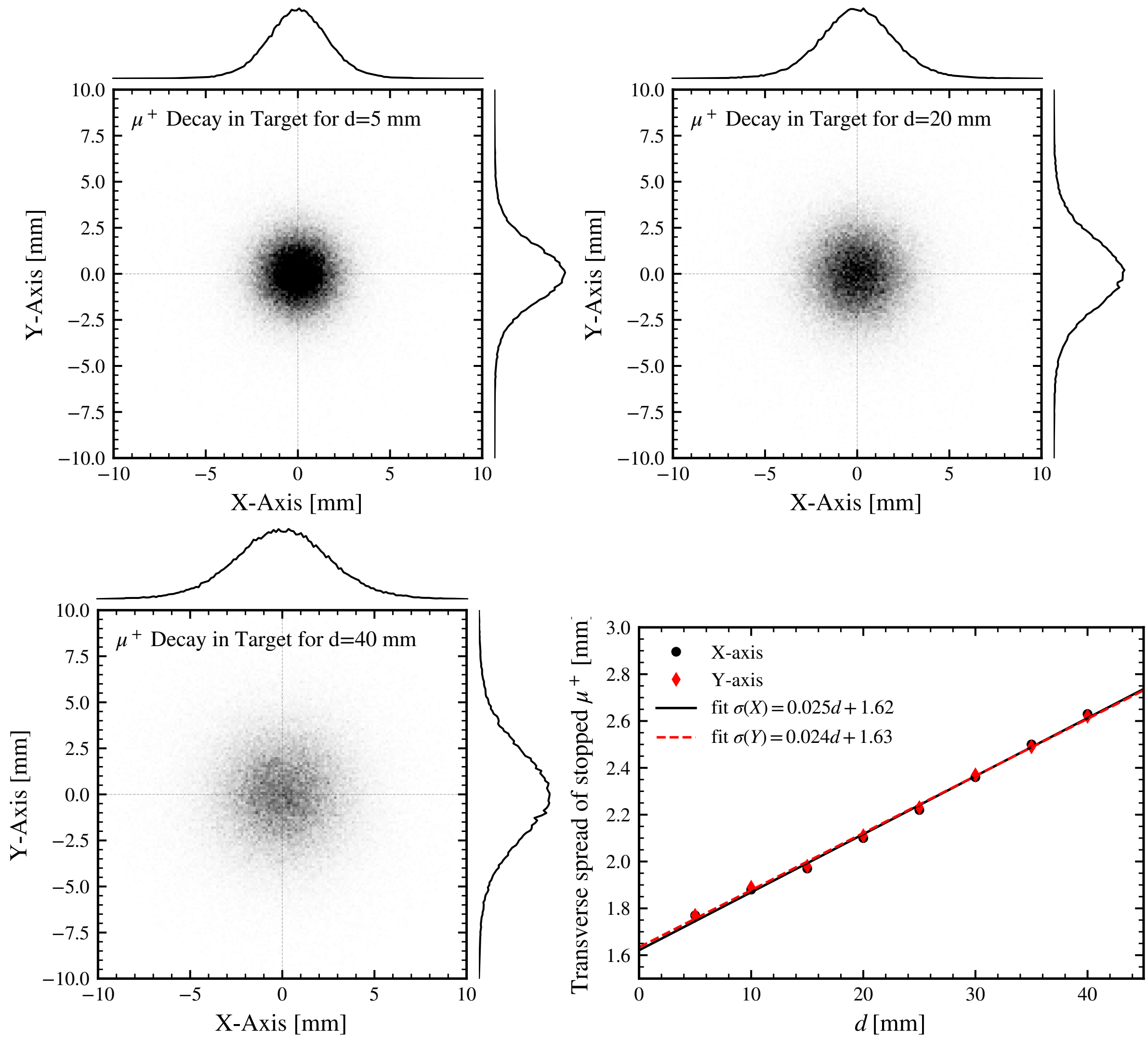


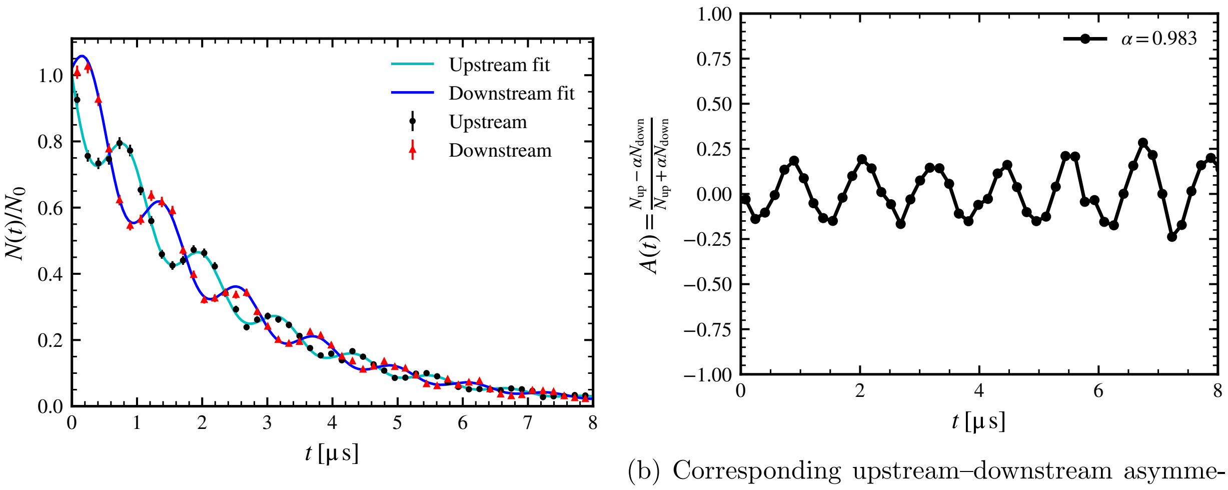
Figure 8: Transverse distribution of stopped muons in the target for selected inner-layer distances d. The lower-right panel shows the standard deviation of the stopped-muon transverse distribution as a function of d. This plot is used as a stopping-distribution sanity check, not as the primary reconstruction residual.

A.2 Simulation 2

This appendix shows the truth-level transverse-field spectrum used as a sanity check before applying the full reconstruction and matching procedure (see Fig. 9).

Table 7: Ideal transverse-field vx-μSR spectra fit-parameters shown in Fig. 9.

Parameter	Upstream	Downstream
N_0 [counts]	2500	2479
A	-0.154	-0.166
ω [rad/μs]	5.349	5.328
ϕ [rad]	-1.567	1.701



(a) Ideal upstream and downstream spectra.

(b) Corresponding upstream–downstream asymmetry.

Figure 9: Truth-level transverse-field μSR signal for a 6 mm-diameter aluminium target in a uniform 6.3 mT magnetic field. The spectra are built directly from simulated decay quantities before applying the full track-reconstruction and matching procedure. The time window is $0 < \Delta t < 13\,\mu s$.